SHETH L.U.J AND SIR M.V COLLEGE
PRACTICAL :-M1

AIM :-

Aim - Practical 1 to 6Generating descriptive statistics using summary() or describe() (R)

OUTPUT :-

```
R 4.1.2 · ~/

> names(df)
[1] "Car_Name"      "Year"          "Selling_Price" "Present_Price" "Kms_Driven"    "Fuel_Type"     "Seller_Type"
[8] "Transmission"  "Owner"
> library(psych)
>
> df <- cardata
>
> print("---- 1. Descriptive Statistics ----")
[1] "---- 1. Descriptive Statistics ----"
>
> print("Summary of a numeric variable:")
[1] "Summary of a numeric variable:"
> summary(df[sapply(df, is.numeric)][,1])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   2003    2012    2014    2014    2016    2018
>
> print("Detailed Description of numeric variables:")
[1] "Detailed Description of numeric variables:"
> describe(df[sapply(df, is.numeric)])
              vars   n     mean       sd  median  trimmed      mad     min      max     range  skew kurtosis      se
Year             1 301  2013.63     2.89  2014.0  2014.00     2.97 2003.00   2018.0     15.00 -1.23     1.46    0.17
Selling_Price    2 301     4.66     5.08     3.6     3.74     3.85    0.10     35.0     34.90  2.47     8.66    0.29
Present_Price    3 301     7.63     8.64     6.4     6.18     6.89    0.32     92.6     92.28  4.04    30.93    0.50
Kms_Driven       4 301 36947.21 38886.88 32000.0 32133.00 25204.20  500.00 500000.0 499500.00  6.37    66.94 2241.40
Owner            5 301     0.04     0.25     0.0     0.00     0.00    0.00      3.0      3.00  7.54    71.59    0.01
> |
```

NAME :- SAURAV KANDARI
ROLL NO :- S084

SHETH L.U.J AND SIR M.V COLLEGE
PRACTICAL :-M1

2) Generating frequency tables using table() or count() (R

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function   Addins

Source

R 4.1.2 · ~/

```
                    1                          3
>
> print("Frequency Table using count():")
[1] "Frequency Table using count():"
> print(df %>% count(.data[[cat_var]]))
                   Car_Name  n
1                       800  1
2                 Activa 3g  2
3                 Activa 4g  1
4                  alto 800  1
5                  alto k10  5
6                     amaze  7
7              Bajaj  ct 100  1
8          Bajaj Avenger 150  1
9   Bajaj Avenger 150 street  1
10         Bajaj Avenger 220  3
11    Bajaj Avenger 220 dtsi  2
12  Bajaj Avenger Street 220  1
13        Bajaj Discover 100  1
14        Bajaj Discover 125  2
15         Bajaj Dominar 400  1
16      Bajaj Pulsar  NS 200  1
17       Bajaj Pulsar 135 LS  1
18          Bajaj Pulsar 150  4
19        Bajaj Pulsar 220 F  2
20       Bajaj Pulsar  NS 200  3
21        Bajaj Pulsar  RS200  1
22                    baleno  1
23                      brio 10
24                     camry  1
25                      ciaz  9
26                      city 26
27                   corolla  1
28             corolla altis 16
29                     creta  3
30                     dzire  4
31                   elantra  2
32                       eon  6
33                    ertiga  6
34               etios cross  3
35                   etios g  3
36                  etios gd  1
37                 etios liva  4
38                  fortuner 11
```

Search

NAME :- SAURAV KANDARI
ROLL NO :- S084

SHETH L.U.J AND SIR M.V COLLEGE
PRACTICAL :-M1

Source

R 4.1.2 · ~/

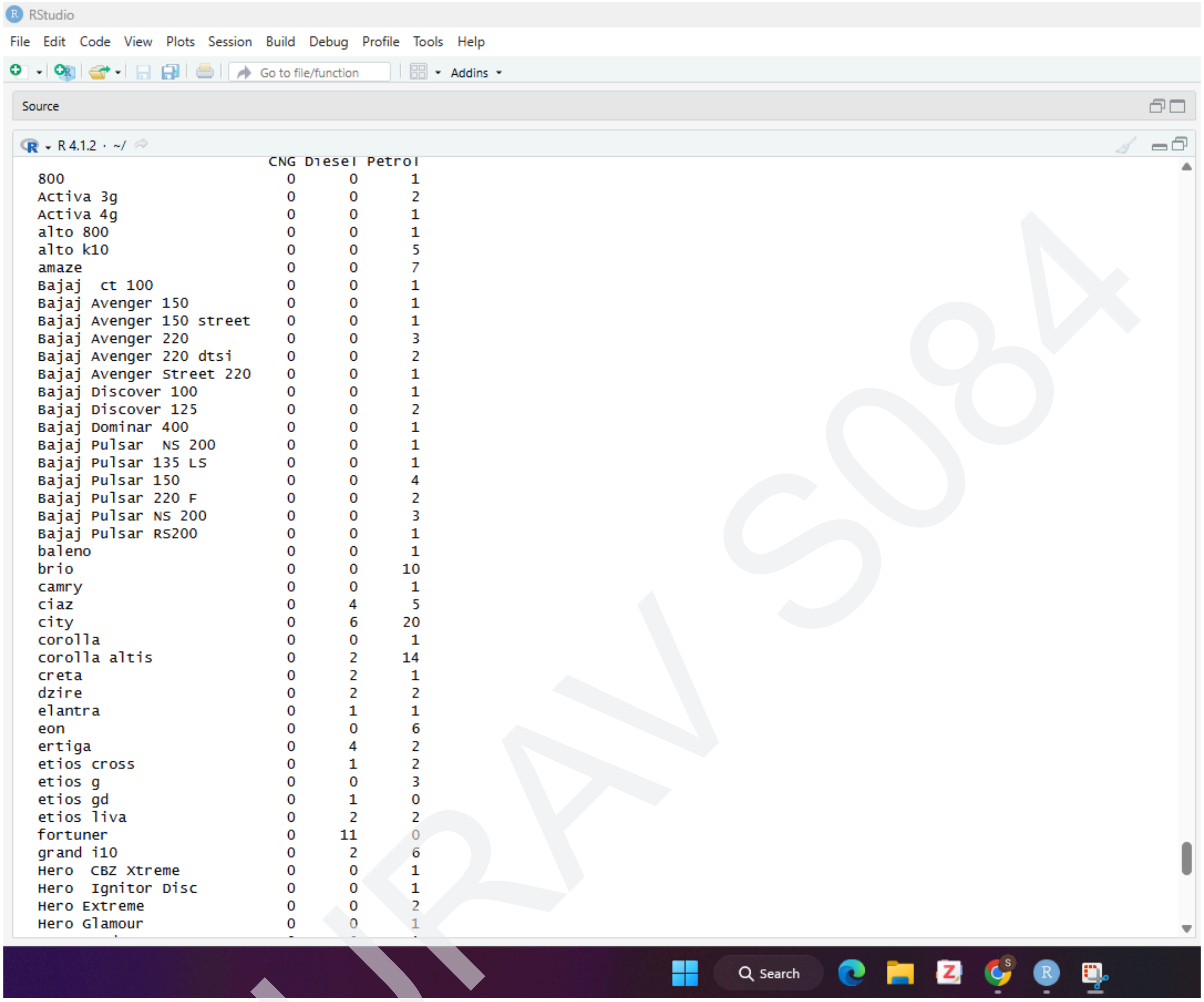
```
51         Hero Super Splendor  1
52            Honda Activa 125  1
53             Honda Activa 4G  2
54         Honda CB Hornet 160R  3
55              Honda CB Shine  2
56            Honda CB Trigger  1
57            Honda CB twister  2
58            Honda CB Unicorn  1
59              Honda CBR 150  2
60             Honda Dream Yuga  1
61               Honda Karizma  2
62              Hyosung GT250R  1
63                         i10  5
64                         i20  9
65                       ignis  1
66                      innova  9
67                        jazz  7
68               KTM 390 Duke   1
69                   KTM RC200  2
70                   KTM RC390  1
71                land cruiser  1
72          Mahindra Mojo XT300  1
73                        omni  1
74                        ritz  4
75    Royal Enfield Bullet 350  1
76   Royal Enfield Classic 350  7
77   Royal Enfield Classic 500  2
78   Royal Enfield Thunder 350  4
79   Royal Enfield Thunder 500  3
80                     s cross  1
81           Suzuki Access 125  1
82                       swift  5
83                         sx4  6
84           TVS Apache RTR 160  3
85           TVS Apache RTR 180  2
86                 TVS Jupyter  1
87                   TVS Sport  1
88                    TVS Wego  1
89           UM Renegade Mojave  1
90                       verna 14
91               vitara brezza  1
92                     wagon r  4
93                      xcent  3
94                Yamaha Fazer  1
95            Yamaha FZ  v 2.0  2
```

Search

NAME :- SAURAV KANDARI
ROLL NO :- S084

SHETH L.U.J AND SIR M.V COLLEGE
PRACTICAL :-M1

3) Creating cross-tabulations and two-way tables using table() (R)




```
                          CNG Diesel Petrol
  800                       0      0      1
  Activa 3g                 0      0      2
  Activa 4g                 0      0      1
  alto 800                  0      0      1
  alto k10                  0      0      5
  amaze                     0      0      7
  Bajaj  ct 100             0      0      1
  Bajaj Avenger 150         0      0      1
  Bajaj Avenger 150 street  0      0      1
  Bajaj Avenger 220         0      0      3
  Bajaj Avenger 220 dtsi    0      0      2
  Bajaj Avenger Street 220  0      0      1
  Bajaj Discover 100        0      0      1
  Bajaj Discover 125        0      0      2
  Bajaj Dominar 400         0      0      1
  Bajaj Pulsar  NS 200      0      0      1
  Bajaj Pulsar 135 LS       0      0      1
  Bajaj Pulsar 150          0      0      4
  Bajaj Pulsar 220 F        0      0      2
  Bajaj Pulsar NS 200       0      0      3
  Bajaj Pulsar RS200        0      0      1
  baleno                    0      0      1
  brio                      0      0     10
  camry                     0      0      1
  ciaz                      0      4      5
  city                      0      6     20
  corolla                   0      0      1
  corolla altis             0      2     14
  creta                     0      2      1
  dzire                     0      2      2
  elantra                   0      1      1
  eon                       0      0      6
  ertiga                    0      4      2
  etios cross               0      1      2
  etios g                   0      0      3
  etios gd                  0      1      0
  etios liva                0      2      2
  fortuner                  0     11      0
  grand i10                 0      2      6
  Hero  CBZ Xtreme          0      0      1
  Hero  Ignitor Disc        0      0      1
  Hero Extreme              0      0      2
  Hero Glamour              0      0      1
```


NAME :- SAURAV KANDARI
ROLL NO :- S084

```
R 4.1.2 · ~/
 Hero Honda CBZ extreme        0      0      1
 Hero Honda Passion Pro        0      0      1
 Hero Hunk                     0      0      1
 Hero Passion Pro              0      0      2
 Hero Passion X pro            0      0      1
 Hero Splender iSmart          0      0      2
 Hero Splender Plus            0      0      1
 Hero Super Splendor           0      0      1
 Honda Activa 125              0      0      1
 Honda Activa 4G               0      0      2
 Honda CB Hornet 160R          0      0      3
 Honda CB Shine                0      0      2
 Honda CB Trigger              0      0      1
 Honda CB twister              0      0      2
 Honda CB Unicorn              0      0      1
 Honda CBR 150                 0      0      2
 Honda Dream Yuga              0      0      1
 Honda Karizma                 0      0      2
 Hyosung GT250R                0      0      1
 i10                           0      0      5
 i20                           0      3      6
 ignis                         0      0      1
 innova                        0      6      3
 jazz                          0      0      7
 KTM 390 Duke                  0      0      1
 KTM RC200                     0      0      2
 KTM RC390                     0      0      1
 land cruiser                  0      1      0
 Mahindra Mojo XT300           0      0      1
 omni                          0      0      1
 ritz                          0      1      3
 Royal Enfield Bullet 350      0      0      1
 Royal Enfield Classic 350     0      0      7
 Royal Enfield Classic 500     0      0      2
 Royal Enfield Thunder 350     0      0      4
 Royal Enfield Thunder 500     0      0      3
 s cross                       0      1      0
 Suzuki Access 125             0      0      1
 swift                         0      2      3
 sx4                           1      1      4
 TVS Apache RTR 160            0      0      3
 TVS Apache RTR 180            0      0      2
 TVS Jupyter                   0      0      1
 TVS Sport                     0      0      1
```

4) Performing one-sample t-tests using t.test() (R)

```
R 4.1.2 · ~/
[1] "---- 4. One-Sample t-test ----"
>
> num_var <- names(df)[sapply(df, is.numeric)][1]
>
> test_value <- 100
>
> t_test_one <- t.test(df[[num_var]], mu = test_value)
>
> print(t_test_one)

        One Sample t-test

data:  df[[num_var]]
t = 11482, df = 300, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 100
95 percent confidence interval:
 2013.300 2013.956
sample estimates:
mean of x
 2013.628
```

SHETH L.U.J AND SIR M.V COLLEGE
PRACTICAL :-M1

5) Performing independent two-sample t-tests using t.test() with grouping (R)

```
>
> num_var <- names(df)[sapply(df, is.numeric)][1]
>
> cat_vars <- names(df)[!sapply(df, is.numeric)]
>
> binary_var <- cat_vars[
+   sapply(cat_vars, function(x) length(unique(df[[x]])) == 2)
+ ][1]
>
> df[[binary_var]] <- as.factor(df[[binary_var]])
>
> t_test_two <- t.test(df[[num_var]] ~ df[[binary_var]])
>
> print(t_test_two)

        Welch Two Sample t-test

data:  df[[num_var]] by df[[binary_var]]
t = 0.65327, df = 184.06, p-value = 0.5144
alternative hypothesis: true difference in means between group Dealer and group Individual is not equal to 0
95 percent confidence interval:
 -0.4870866  0.9693314
sample estimates:
    mean in group Dealer mean in group Individual
                2013.713                 2013.472

> |
```

6) Performing paired t-tests using t.test(paired=TRUE) (R)

```
R 4.1.2 · ~/
> num_var <- names(df)[sapply(df, is.numeric)][1]
>
> set.seed(123)
>
> before_values <- df[[num_var]] - runif(nrow(df), min = 0.1, max = 0.5)
> after_values  <- df[[num_var]]
>
> t_test_paired <- t.test(
+   before_values,
+   after_values,
+   paired = TRUE
+ )
>
> print(t_test_paired)

        Paired t-test

data:  before_values and after_values
t = -46.401, df = 300, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.3129561 -0.2874908
sample estimates:
mean of the differences
             -0.3002234

> |
```


NAME :- SAURAV KANDARI
ROLL NO :- S084